

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination

May 2012

CL102 Environmental Sciences

Date: 09.05.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Define the term Environment. Briefly describe physical components of it. [03]
(b) Give the composition of atmosphere in brief. [04]
- Q - 2 (a) Enlist components of Environment and explain lithosphere in detail. [04]
(b) Define Noise Pollution. Describe sources of noise pollution. [05]
(c) Enlist parameters of water quality standards for drinking water in India specified by Indian Standards. [05]

OR

- Q - 2 (a) Describe Structure of Atmosphere in detail. [04]
(b) Explain the term "Environmental Degradation" and discuss the role of advanced Technology in the degradation of environment. [05]
(c) Describe effects of Air Pollution on plants. [05]
- Q - 3 (a) Give the classification of air pollutants based on source. [04]
(b) Enlist and explain any two sources of water pollution. [05]
(c) What is 'Acid Rain'? Write a brief note on effects of acid rain on environment. [05]

OR

- Q - 3 (a) Enlist the common air pollutants stating their permissible limits as per ambient air quality standards. [04]
(b) Enlist all current global environmental problems and write a note on 'Green House Effect'. [05]
(c) Give the classification of water quality parameters and explain any two. [05]

SECTION - II

- Q - 4 (a) Give examples of organic and inorganic components in ecosystem. [02]
(b) Define ecosystem and enlist various major ecosystem. [02]
(c) Describe food web with example. [03]

- Q - 5 (a) Discuss how energy flow differs from nutrient flow in an ecosystem. [04]
 (b) Define Biogeochemical cycle. Write environmental significance of nitrogen cycle. [05]
 (c) Define autoecology. Give advantages and limitations of solar energy [05]

OR

- Q - 5 (a) What do you understand by structure of ecosystem? Give classification of consumers. [04]
 (b) Enlist types of food chain in an ecosystem. Explain ecological pyramid of energy. [05]
 (c) Enlist components of forest ecosystem. Enlist impacts due to overexploitation of water resources. [05]

- Q - 6 (a) Explain Demographic transition theory. [04]
 (b) Enlist methods of population forecasting and explain how overpopulation is acting as pollutants. [05]
 (c) Describe the environmental impacts due to overexploitation of the forest resources. [05]

OR

- Q - 6 (a) Which are the important theories of population growth? Explain Malthusian theory. [04]
 (b) How is habitation pattern developed? List all environmental factors governing human settlement. [05]
 (c) Give classification of natural resources. Why do we conserve them? [05]
